

Tianzhi Li

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Homepage: <https://tianzhi-li.github.io>

Basic Information

I am final-year PhD student at Peking University (PKU) working with Professors [Zhisheng Duan](#) and [Jinzhi Wang](#). From Dec-2024 to Nov-2025, I was fortunate to work as a visiting PhD at Nanyang Technological University (NTU), Singapore, with Professor [François Gay-Balmaz](#). Prior to that, I received my B.Sc. in Mathematics from Beijing Institute of Technology (BIT) in 2021 under the supervision of Professor [Donghua Shi](#).

My research interest lies at the intersection of **Geometric Mechanics & Control**, **Physics-Informed Learning**, and **Stochastic Dynamics** for robotic and mechanical systems. I am broadly interested in leveraging physical principles and differential geometry to develop computationally efficient algorithms for **control**, **estimation**, and **learning** with sound theoretical guarantees.

Academic Experience

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|---|---------------------|
| ◦ Peking University
PhD (Dynamical Systems and Control) | Sep 2021 - Jun 2026 |
| ◦ Nanyang Technological University , Singapore
Visiting PhD (Mechanics and Control) | Dec 2024 - Nov 2025 |
| ◦ Beijing Institute of Technology
Bachelor of Science (Mathematics) | Sep 2017 - Jun 2021 |

Teaching Experience

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| - Teaching Assistant, “Ordinary Differential Equations”, Peking University
Undergraduate-level course, covering more than 150 students
Recieved Outstanding Teaching Assistant Award | Fall 2023 |
| - Teaching Assistant, “Linear Algebra”, Peking University
Undergraduate-level course, covering more than 150 students | Spring 2023 |
| - Teaching Assistant, “Linear Algebra”, Peking University
Undergraduate-level course, covering more than 150 students
Delivered Lecture 5 “Determinants”
Recieved Outstanding Teaching Assistant Award | Fall 2022 |
| - Teaching Assistant, “Analytic Mechanics”, Peking University
Undergraduate-level course, covering more than 100 students | Spring 2022 |
| - Teaching Assistant, “Ordinary Differential Equations”, Peking University
Undergraduate-level course, covering more than 150 students | Fall 2021 |

Publications

- [1] **T. Li** and J. Wang, Variational Unscented Kalman Filter on Matrix Lie Groups, **Automatica** (Regular Paper), 172: 111995, 2025. [[Paper Link](#)]
- [2] **T. Li**, R. Fu, and J. Wang, Reduced Dynamics and Geometric Optimal Control of Nonequilibrium Thermodynamics: Gaussian Case, **Automatica** (Regular Paper), 164: 111626, 2024. [[Paper Link](#)]
- [3] **T. Li** and J. Wang, Physics-Informed Gaussian Process Learning on Lie Groups, **Journal of Guidance, Control, and Dynamics**, 48 (11), pp. 2654-2662, 2025. [[Paper Link](#)]
- [4] **T. Li**, J. Wang, and Z. Duan, Structure-Preserving Unscented Kalman Filter for Planar Mobile Robots, **IEEE Control Systems Letters**, vol. 9, pp. 2157-2162, 2025. [[Paper Link](#)]
- [5] **T. Li**, F. Gay-Balmaz, D. Shi, and J. Wang, Variational Principle for Stochastic Nonholonomic

Systems **Part II**: Stochastic Nonholonomic Integrator. In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information (**GSI'25**), Saint-Malo, France, vol. 16034, pp. 225-233. Springer, 2026. [[Paper Link](#)]

- [6] **T. Li**, F. Gay-Balmaz, D. Shi, and J. Wang, Variational Principle for Stochastic Nonholonomic Systems **Part I**: Continuous-Time Formulation. In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information (**GSI'25**), France, vol. 16034, pp. 204-213. Springer, 2026. [[Paper Link](#)]
- [7] **T. Li** and J. Wang, A Structure-Preserving Learning Scheme on SO(3), 2024 43rd IEEE Chinese Control Conference (**CCC'24**), Kunming, China, 2024, pp. 5149-5152. [[Paper Link](#)]
- [8] **T. Li** and J. Wang, Multisymplectic Unscented Kalman Filter for Geometrically Exact Beams. In: Nielsen, F., Barbaresco, F. (eds) International Conference on Geometric Science of Information (**GSI'23**). Lecture Notes in Computer Science, Saint-Malo, France, vol. 14072, pp. 60-68, Springer Verlag. [[Paper Link](#)]
- [9] **T. Li** and J. Wang, A Physics-Informed Gaussian Process Regression Algorithm for The Dynamics of The Planar Pendulum, 2023 42nd IEEE Chinese Control Conference (**CCC'23**), Tianjin, China, pp. 5163-5167, 2023. [[Paper Link](#)]
- [10] **T. Li** and J. Wang, A Statistical Dynamical Algorithm for Gaussian Multi-Agent Systems Under Hamel's Formalism, 34th IEEE Chinese Control and Decision Conference (**CCDC'22**), Hefei, China, pp. 1344-1349, 2022. [[Paper Link](#)]
- [11] M. Du, F. Gay-Balmaz, **T. Li**, and D. Shi, Stochastic Geometrically Exact Beam: Continuous-Time Geometric Principles and Variational Discretizations, To be submitted, 2026.

Honors and Awards

◦ National Scholarship (top 1%) Issued by Ministry of Education, China	Sep-2025
◦ Merit Student of Beijing Issued by Beijing Education Commission	Dec-2025
◦ Presidential Doctoral Scholarship (The most prestigious scholarship issued by Peking University)	Jun-2025
◦ China Scholarship Council Visiting PhD Scholarship Issued by China Scholarship Council	Jul-2024
◦ Merit Student Award (top 1%) Issued by Peking University	Sep-2025
◦ Excellent Academic Research Award (top 1%) (only 2 awardees at SAMR, Peking University)	Sep-2025
◦ Dean's Doctoral Scholarship - First Prize Issued by College of Engineering, Peking University	Oct-2024
◦ Yuehua Luo Scholarship Issued by Peking University	Nov-2024
◦ Outstanding Teaching Assistant Award (× 2) Issued by College of Engineering, Peking University	Mar-2024 & Apr-2023

Conference Talks & Invited Talks

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- Variational Principle for Stochastic Nonholonomic Systems Part II: Stochastic Nonholonomic Integrator
The International Conference on Geometric Science of Information (GSI)
Saint-Malo, France, Oct-2025
 - Variational Principle for Stochastic Nonholonomic Systems Part I: Continuous-Time Formulation
The International Conference on Geometric Science of Information (GSI)

Saint-Malo, France, Oct-2025

- A Structure-Preserving Learning Scheme on $SO(3)$
The 43rd Chinese Control Conference (CCC)
Kunming, China, Jul-2024
- Multisymplectic Unscented Kalman Filter for Geometrically Exact Beams
International Conference on Geometric Science of Information (GSI)
Saint-Malo, France, Aug-2023
- A Physics-Informed Gaussian Process Regression Algorithm for the Dynamics of the Planar Pendulum
The 42nd Chinese Control Conference (CCC)
Tianjin, China, Jul-2023
- A Statistical Dynamical Algorithm for the Dynamics of Gaussian Distributions
The 34th Chinese Control and Decision Conference (CCDC)
Hefei, China, Aug-2022
- Geometric Structures and Optimal Control of Gaussian Distributions
Workshop on Multibody System Dynamics, School of Aerospace Engineering, Beijing Institute of Technology, Apr-2024. (Invited by Profs. Qiang Tian and Ju Chen)
- Variational Unscented Kalman Filter on Matrix Lie Groups (Invited by Prof. Donghua Shi)
School of Mathematics and Statistics, Beijing Institute of Technology, Jul-2024.

Academic Service

Journal Reviewer: Automatica, International Journal of Dynamics and Control.

Conference Reviewer: IEEE Conference on Decision and Control (CDC), IFAC Workshop on Lagrangian and Hamiltonian Methods for Nonlinear Control (LHMNC), International Symposium on Mathematical Theory of Networks and Systems (MTNS).

References

Professor Zhisheng Duan

Position: Professor of Dynamics and Control

Deputy Director of State Key Laboratory for Turbulence & Complex Systems

Affiliation: Peking University

E-mail: duanzs@pku.edu.cn

Professor François Gay-Balmaz

Position: Associate Professor of Mathematics

Affiliation: Nanyang Technological University

E-mail: francois.gb@ntu.edu.sg

Professor Jinzhi Wang

Position: Professor of Dynamics and Control

Affiliation: Peking University

E-mail: jinzhiw@pku.edu.cn

Professor Donghua Shi

Position: Professor of Mathematics

Director of Beijing Key Laboratory on MCAACI

Affiliation: Beijing Institute of Technology

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